

Exact Nested-Sphere Geometry and the Breathing Dimension-Counter:

S^3, S^4, S^5 , the $+\ell$ Cascade, and the 3:4:5 Triad

TZPID Gold Spine Series II, Paper XIII of XX

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June 2026

Abstract

The breathing enclosure favoured by Paper XII demands an exact geometry. We give the dimension-general geometry of the 3-, 4-, and 5-spheres as nested breathing enclosures: closed-form hypervolumes $|S^n|R^n$, the breathing law $\dot{V}/V = nH$, the Laplace–Beltrami spectrum $\lambda_\ell = \ell(\ell + n - 1)/R^2$ with exact degeneracies, scalar curvature $n(n - 1)/R^2$, and the higher-dimensional Friedmann scaling. Two exact structural results emerge: the nesting rule $\lambda_\ell(S^n) = \lambda_\ell(S^{n-1}) + \ell$, the cascade-plus-cutoff mechanism of Paper I made exact; and the Pythagorean fundamental triad $\lambda_1(S^3) : \lambda_1(S^4) : \lambda_1(S^5) = 3 : 4 : 5$, tying the nested fundamental tones to the 3–4–5 right triangle and, through the shared S^1 Hopf fiber, to the comma holonomy of Paper III. The load-bearing result is a measurement: because $\dot{V}/V = nH$ exactly, the ratio of cosmic volume-production rate to the Hubble rate *is* the spatial dimension, and the value from the Spartan Dawn fit, $\dot{V}/V = 0.207 \text{ Gyr}^{-1}$, $H_0 \approx 0.069 \text{ Gyr}^{-1}$, gives $n = 3.0$. The observable spatial slice is the 3-sphere; S^4, S^5 are outer shells of the nest. Sections 2–3 are exact closed-form geometry; the dimension anchor is data-driven; the identification of nature with the stack $S^3 \subset S^4 \subset S^5$ remains the hypothesis tested in Paper XIV.

1 Introduction: the geometry the data demand

Paper XII left the breathing enclosure favoured but under-specified. This paper supplies the structure of the nest in closed form, treating the nested hyperspheres as exact Riemannian objects. The computation is standard differential geometry; what the framework adds is the reading of these quantities as the physical content of nested cosmological enclosures, and the recognition that one of them is directly measured.

2 The breathing dimension-counter

Volume on an n -sphere of radius R is $V = |S^n|R^n$, so

$$\boxed{\frac{\dot{V}}{V} = n \frac{\dot{R}}{R} = nH} \tag{1}$$

exactly. The ratio of volume-production rate to the Hubble rate is literally the number of spatial dimensions. The Spartan Dawn fit supplies both numbers independently: $\dot{V}/V = 0.207 \text{ Gyr}^{-1}$, $H_0 \approx 0.069 \text{ Gyr}^{-1}$, whence $n = 0.207/0.069 = 3.0$.

Theorem 2.1 (Dimension anchor). *The observable spatial slice is the 3-sphere. Consequently S^4 and S^5 are not extra directions of measured space; they are outer shells of the nested enclosure, geometric and structural layers rather than observed spatial dimensions.*

Were space S^4 , Eq. (1) would give $\dot{V}/V = 0.28 \text{ Gyr}^{-1}$; were it S^5 , 0.35 Gyr^{-1} . The measured $3H_0$ fixes the factor 3 in ID10867 as a measurement of dimensionality.

3 The three spheres, exactly

The unit volume is $|S^n| = 2\pi^{(n+1)/2}/\Gamma(\frac{n+1}{2})$; the eigenvalues $\lambda_\ell = \ell(\ell + n - 1)$; the degeneracies $m_\ell = \binom{n+\ell}{\ell} - \binom{n+\ell-2}{\ell-2}$; the scalar curvature $n(n-1)/R^2$. The eigenvalue ladders at unit radius are 0, 3, 8, 15, 24, ...

Table 1: Exact dimension-general geometry of the nested spheres.			
Quantity	S^3 (our space)	S^4	S^5
Unit volume $ S^n $	$2\pi^2$	$\frac{8}{3}\pi^2$	π^3
Volume at radius R	$2\pi^2 R^3$	$\frac{8}{3}\pi^2 R^4$	$\pi^3 R^5$
Breathing $\dot{V}/V$	$3H$	$4H$	$5H$
Scalar curvature	$6/R^2$	$12/R^2$	$20/R^2$
Spectrum λ_ℓ	$\ell(\ell+2)/R^2$	$\ell(\ell+3)/R^2$	$\ell(\ell+4)/R^2$
Fundamental gap λ_1	$3/R^2$	$4/R^2$	$5/R^2$
Degeneracy m_ℓ	$(\ell+1)^2$	$\frac{(\ell+1)(\ell+2)(2\ell+3)}{6}$	$\frac{(\ell+1)(\ell+2)^2(\ell+3)}{12}$
m_ℓ for $\ell = 0..4$	1, 4, 9, 16, 25	1, 5, 14, 30, 55	1, 6, 20, 50, 105
Density of states	$\sim \ell^2$	$\sim \ell^3$	$\sim \ell^4$

(S^3); 0, 4, 10, 18, 28, ... (S^4); 0, 5, 12, 21, 32, ... (S^5).

4 The nesting rule and the Pythagorean triad

Proposition 4.1 (The $+\ell$ cascade). *Stepping outward one shell raises every mode by its own level number:*

$$\boxed{\lambda_\ell(S^n) = \lambda_\ell(S^{n-1}) + \ell}, \quad \ell(\ell + n - 1) - \ell(\ell + n - 2) = \ell. \quad (2)$$

Low modes barely move; high modes climb fast, so the nested shells fan apart at the top of the ladder.

This is the exact form of the cascade-plus-cutoff mechanism Paper I exhibited across sand, water, and cloud ripples: each shell's spectrum is the previous shell's sheared upward, linearly in the mode index.

Proposition 4.2 (The fundamental triad). *The fundamental gaps stand in a Pythagorean ratio,*

$$\boxed{\lambda_1(S^3) : \lambda_1(S^4) : \lambda_1(S^5) = 3 : 4 : 5}. \quad (3)$$

The breathing dimension and the first overtone ($\lambda_1 = n/R^2$) are the same integer. Both S^3 and S^5 are circle bundles ($S^3 \rightarrow S^2$ and $S^5 \rightarrow \mathbb{CP}^2$, fiber S^1), so the same S^1 phase that carries the comma holonomy of Paper III threads the nested shells.

5 Higher-dimensional Friedmann scaling

The generalized breathing law in n spatial dimensions is $H^2 = \frac{16\pi G}{n(n-1)}\rho - kc^2/R^2$, $\dot{\rho} + nH(\rho + p/c^2) = 0$. Matter dilutes as R^{-n} , radiation as $R^{-(n+1)}$, vacuum stays constant, curvature as R^{-2} . Two invariants survive every dimension: vacuum density is constant (the $p = -\rho c^2$ fixed point of Papers XV–XX), and curvature always dilutes as R^{-2} —the geometric root of the permitted-but-unconstrained Ω_K of Paper XII. The radiation exponent $n + 1 = 4$ for S^3 is the quartic that fixes the dark-energy slope in Paper XX.

6 Demarcation

Established mathematics: the hypervolumes, Laplace spectra, degeneracies, scalar curvatures, the nesting identity, the 3:4:5 gap ratio, and the generalized Friedmann scaling are exact and standard.

Framework synthesis: the reading of these spheres as physical nested cosmological enclosures; $\dot{V}/V = 3H$ as a dimension measurement (ID10867); and the linkage of the triad to the comma spine through the shared S^1 Hopf fiber.

Predictions: a future precision determination of $\dot{V}/V$ departing from $3H_0$ would contradict $n = 3$; and any observable trace of S^4 , S^5 is deferred to Paper XIV, where it is permitted only through a fine-tuned channel.

7 Limits of the construction

The exactness of Sections 2–3 must not be over-read. $n = 3$ is data-anchored, not derived. That S^4 and S^5 are *physical* outer shells rather than convenient higher embeddings is a hypothesis; there is no observational evidence for extra large spatial dimensions, and this paper claims none. Their role is structural: a nested mode budget whose cascade projects into the 3-sphere. The 3:4:5 triad and comma linkage are exact arithmetic, but their physical operativeness is the framework wager, tested acoustically in Paper XIV.

8 Conclusion: three is measured, the rest is structure

The breathing enclosure now has an exact geometry: closed-form volumes, spectra, degeneracies, curvatures; shells nesting by $\lambda_\ell(S^n) = \lambda_\ell(S^{n-1}) + \ell$; fundamental tones in the 3–4–5 triad; and a breathing rate that counts the spatial dimension, fixed at three. Paper XIV feeds the outer-shell ladders through the BAO, the one observable that could carry their fingerprint.

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